

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester I Examination, 2016-17 (Old Course)

Paper No. MTB-102: Calculus

Time: 3 Hours

Full Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 2 marks each.)

- (a) Give an example of a sequence which is bounded but not convergent.
- (b) By using $\epsilon - \delta$ definition, prove that $\lim_{x \rightarrow 2}(x^2 + 2) = 6$.
- (c) Show that the function defined by

$$f(x) = \begin{cases} \frac{|x-a|}{x-a}, & x \neq a \\ 1, & x = a \end{cases}$$

is not continuous at $x = a$. Write the type of discontinuity at $x = a$.

- (d) If $2^x + 2^y = 2^{x+y}$, then find the value of $\frac{dy}{dx}$.
- (e) If $y = \sin^2 x$, then find y_n .
- (f) If $\cos x = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots$, then find the value of a_3 .
- (g) Expand $f(x) = 2x^3 + 5x^2 + x - 1$ in powers of $(x - 1)$.
- (h) Verify Rolle's theorem for the function $f(x) = x^2 - 4x + 3$ on $[1, 3]$.
- (i) Find all asymptotes of the curve $xy - y = 0$.
- (j) Find the nature of the multiple point $(0, 0)$ on the curve $x^3 + y^3 - 3axy = 0$.
- (k) Find the value of $\lim_{n \rightarrow \infty} \left[\frac{1}{n} + \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} \right]$.

Question 2.

- (a) Let $\dots \lim_{x \rightarrow a} f(x) = l$ and $\lim_{x \rightarrow a} g(x) = m$. Prove that $\lim_{x \rightarrow a} f(x)g(x) = lm$. [7]
- (b) Examine the continuity of the function $f(x) = [x]$, where $[x]$ is the greatest integer function, at $x = 0.5$ and $x = 1$. [5]

Question 3.

- (a) State and prove Lagrange's mean value theorem. Also write its geometrical meaning. [6]

- (b) Prove that for all finite values of x :

$$e^x \sin x = x + x^2 + \frac{2x^3}{3!} - \frac{4x^5}{5!} + \dots$$

[6]

Question 4.

- (a) Show that every differentiable function is continuous, but the converse is not true. [5]

- (b) Test the differentiability of the function

$$f(x) = \begin{cases} 2 + x, & x < 1 \\ 4 - x, & x \geq 1 \end{cases}$$

at $x = 1$.

[4]

- (c) By using Cauchy's mean value theorem, show that:

$$\frac{\sin \beta - \sin \alpha}{\cos \alpha - \cos \beta} = \cot \theta, \quad 0 < \alpha < \theta < \beta < \frac{\pi}{2}$$

[3]

Question 5.

- (a) State and prove Leibniz's theorem. [5]

- (b) If $y = \cos(m \sin^{-1} x)$, $m \in \mathbb{N}$, then show that:

$$(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} + (m^2 - n^2)y_n = 0$$

and hence find $y_n(0)$.

[7]

Question 6.

- (a) Find all the asymptotes of the curve:

$$x^3 + 2x^2y - xy^2 - 2y^3 + xy - y^2 + 1 = 0$$

[5]

- (b) Trace the curve $y(4 - x^2) = 4x$.

[7]

Question 7.

- (a) Trace the curve $r = a(1 + \cos \theta)$, $a > 0$. [6]

- (b) By using the definition of definite integral as the limit of a sum, evaluate $\int_a^b \cos x \, dx$. [6]